

When Cautious Template Memory Hurts: A Reproducible Training-Free Study of Frozen FARTrackSparse

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Abstract

Template updates make visual trackers adapt to changing appearance, but can also propagate a localization error. We conduct a controlled, training-free study of five memory interventions for a frozen FARTrackSparse checkpoint. The development protocol uses the public GOT-10k validation set (180 sequences, 21,007 frames), while full LaSOT Testing is held out until configuration freezing. Three complete development evaluations show that hard write holding, counterfactual write rejection, and stable-anchor recovery reduce mean frame IoU by 0.00823, 0.02000, and 0.03204, respectively, relative to the immutable baseline. Two soft rebinding variants fail fixed-screen directionality or efficiency gates. Thus no proposed intervention is claimed as an improvement. The contribution of this draft is a transparent negative-result record, including checked implementation failures, statistical uncertainty, and a reproducible protocol. The held-out baseline-only LaSOT run is in progress at the time of this draft and its result is intentionally not reported here.

1 Introduction

Autoregressive tracking exposes a practical tension: recent templates improve adaptation, while an erroneous crop may contaminate later predictions. FARTrack uses autoregression and inter-frame sparsification to obtain fast tracking [3]. We asked a deliberately narrow question: can untrained control of its inference-time template memory improve a released, frozen FARTrackSparse checkpoint without additional training?

This paper reports the answer for the mechanisms tested: no. We regard this as useful evidence because plausible memory protections can silently damage the update distribution expected by a frozen tracker. We do not replace this negative result with a post-hoc positive claim.

2 Protocol

Model and invariants. Every experiment uses the released `FARTrackSparse_ep0015.pth.tar` checkpoint, the original prediction decoder, the original result writer, and the same initialization boxes. The baseline tracker is exposed as `fartrack_sparse_research`; its upstream inference implementation is not modified.

Development/test separation. B_{dev} is the public GOT-10k validation split. GOT-10k was introduced as a high-diversity generic tracking benchmark with an explicit one-shot protocol [2]. The development metric is mean frame IoU (AO) over 180 sequences and 21,007 frames, calculated by a read-only local script which checks sequence coverage and prediction lengths. Full official LaSOT Testing is B_{test} . LaSOT supplies dense, long sequences for SOT evaluation [1]. It was checksum-verified and not accessed until all candidate decisions were frozen. The B_{test} run is baseline-only because no candidate survived B_{dev} .

Statistics and speed. We report the mean of per-frame IoUs. For a candidate-baseline comparison, we resample the 180 per-sequence AO differences 10,000 times with seed 20260819 and report a percentile 95% interval. Aggregate FPS is total frames divided by the sum of saved per-frame times. These intervals describe the observed development set; they are not independent-video population inference.

3 Interventions

1. **TRM-FAR**: a transaction ledger holds writes under three causal anomaly signals and rolls back after sustained failure.
2. **Counterfactual agreement**: old-only and recent-only template views gate a write when their coordinate predictions disagree. It requires two extra forwards after warm-up.
3. **Stable-anchor recovery**: persistent failures temporarily replay the initial template and the last accepted view.
4. **Recent-consensus rebinding**: all frames are written, but fixed template slots are rebound using untrained RGB descriptor medoids.
5. **Appearance-diverse rebinding**: all frames are written, while slots retain chronological max-min RGB descriptor exemplars.

All implementations are separate tracker entry points. Focused unit tests and GPU smoke tests preceded evaluation. Counterfactual agreement initially exposed an out-of-range indexing defect after a rejected write; it was repaired by indexing its bounded template pool by accepted writes, and only the repaired version was evaluated.

4 Results

Table 1: Complete B_{dev} results. AO is mean frame IoU.

Method	AO	Δ AO	95% paired interval	FPS
Immutable FARTrackSparse	0.832741	0.000000	–	40.07
TRM-FAR	0.822471	-0.008228	[-0.019499, 0.004104]	42.16
Counterfactual agreement	0.806470	-0.019995	[-0.037073, -0.003256]	23.47
Stable-anchor recovery	0.797473	-0.032038	[-0.050311, -0.015034]	40.90

TRM-FAR improved 70 sequences and degraded 110; stable-anchor recovery improved 66 and degraded 114; counterfactual agreement improved 77 and degraded 103. No candidate in Table 1 was tuned after its complete development result or run on B_{test} .

The two soft alternatives did not pass their prespecified focused-screen gates. Recent-consensus rebinding obtained 0.907830 versus 0.909803 for the baseline on a fixed 60-frame screen. Appearance-diverse rebinding obtained a small 0.000838 AO increase on a fixed 80-frame screen, but incurred a 16.57% FPS loss (21.93 versus 26.28 FPS), exceeding the 5% gate. These are screening observations, not generalization estimates, and were not expanded to a full development evaluation.

5 Discussion and Limitations

The shared pattern is that hard intervention can sacrifice normal appearance adaptation more often than it prevents memory pollution. Uncertainty, motion inconsistency, and disagreement are not calibrated error probabilities for a frozen model. Further, changing the effective template distribution may create an input regime unseen during training. This study does not establish that template control is never useful; it only rejects these implementations and fixed settings on the stated protocol.

Limitations are substantial. There is one released checkpoint, one full public development split, RGB descriptors rather than learned confidence, and no positive method. The LaSOT test score is pending and must not be inferred from GOT-10k validation. GOT-10k test server submission has not been performed.

6 Reproducibility and Disclosure

All source changes, commands, structured metrics, logs, and immutable checkpoint path are documented in `research/EXPERIMENT_LOG.md`. Data archives and outputs reside on the data disk. This draft was prepared with AI assistance; human authors must verify all text, select authorship, and approve any submission. No funding, conflict-of-interest, or authorship statements are asserted because they were not provided.

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